Mining & Industrial Development Expansion

Owner-approved design specification | Proposed dedicated Build 17 or 18 | 19 October 2026

DESIGN STATUS: Approved direction and scope. Exact mechanics, costs, balancing values, map technology and build number remain to be designed and validated.

1. Vision and purpose

This is not simply a mining update. It is a village-wide industrial development expansion. Mining drives demand for carpenters, forges, haulers, builders, quarry workers and new specialists. As settlements acquire skills, tools, infrastructure and institutional knowledge, villages can develop into towns and eventually major settlements. The player is not required for this progression: capable NPC communities should pursue it autonomously when circumstances support it.

Core principle: a particular mineral deposit is finite, but the mining industry can continue through prospecting, investment, excavation, transport, trade and specialization. Never create ore or wealth from nothing merely to preserve the economy.

2. Five-stage mining development pathway

Stage 1 — Surface deposits

The existing mineral-bearing rock is the visible top deposit. Colored ore indicators mark extractable mineral. When the colored indicators are depleted, the remaining gray rock can still be mined for ordinary stone. Depleted ore does not automatically replenish. Stone is moved into a quarry/stone-handling production chain.

Stage 2 — Prospecting and assay

Prospectors interpret subtle natural clues, investigate nearby terrain with specialized equipment, collect samples and have findings evaluated by an assayer or equivalent specialist. A clue indicates a search area, not a guaranteed deposit or its exact location. Newly located deposits remain finite.

Stage 3 — Open-pit / strip mining

Workers remove topsoil, dirt and rock to excavate a roughly square pit with terraced sides and a descending spiral ramp or switchback access. Deeper strata may reveal new finite deposits. Excavation creates real spoil, access, hauling and safety requirements.

Stage 4 — Underground mines and caves

Suitable rock faces and underground formations may support shafts, galleries, chambers and enterable caves. In isometric view, roofs/overhead terrain hide or become transparent so the player can see characters inside. Supports, lighting, ventilation, access, navigation and hauling are genuine dependencies.

Stage 5 — Mature mining and industrial networks

Advanced settlements develop dedicated quarry and mining operations, minecarts, track laying, track fabrication, cart assembly, repair, hauling logistics, storage, assay services and trade links. Expansion is supported by an actual workforce and resource economy rather than a simple unlock button.

3. Settlement capability levels and growth

Mining capability is a property of the settlement as a functioning community, not just the skill of one NPC. A village can learn to prospect, build specialized infrastructure and train workers; that capability enables greater economic complexity and contributes to its evolution into a town or major settlement.

Level	Capability	Evidence required in world
1	Surface extraction	Access to exposed deposits, miners, tools and working transport
2	Prospecting and assay	Trained prospectors, toolkit, samples, assaying capability and recorded discoveries
3	Open-pit excavation	Earthmoving crews, excavation plans, ramps, spoil handling, carts and ongoing maintenance
4	Underground mining	Shaft/tunnel access, timber supports, lighting, specialist crews and safe transport
5 (proposed)	Integrated industrial mining	Minecart/track industry, specialists, repair, distribution, trade and coordinated operations

Levels 1–4 are the owner's stated progression. Level 5 is a proposed mature-industry extension for later confirmation. Settlement size (village, town, major settlement) and mining capability should be related but not identical: a large trading town may lack mines, while a small specialist village may be skilled at prospecting. Advancement requires real prerequisites and can regress if workers, equipment or knowledge disappear.

4. Prospecting: reading the land

Geological hints should be natural and discoverable: faintly stained or discolored soil, mineral flecks, exposed veins, unusual rock fragments or changes in nearby outcrops. The owner specifically proposes a ground stain of a different color as an initial example. These are hints, not an X marking a guaranteed deposit.

- The terrain generator places finite deposits and plausible surface evidence in proximity, with uncertainty and occasional misleading clues.
- Prospectors examine terrain, choose search locations, use a prospecting toolkit, collect physical samples and carry them to an assayer.
- An assay can estimate mineral type and quality with uncertainty; better tools and experience improve reliability, not omniscience.
- Discoveries are recorded and become actionable knowledge only for people or institutions that receive the information.
- The future large map must provide further geological opportunity without assuming the current map is already large.

5. New careers and the industrial supply chain

Specialist professions

- Prospector: surveys terrain, recognizes clues, samples, marks probable sites and reports discoveries.
- Assayer: tests samples and evaluates mineral composition or quality using tools, fuel and materials as appropriate.
- Miner and excavation crew: extract surface ore, stone, earth and deeper deposits.
- Quarry worker / stonecutter: extracts, sorts and prepares usable stone.
- Mine engineer or foreman (proposed role): plans safe ramps, shafts, supports and excavation work.
- Track layer: installs and maintains mine tracks and associated earthworks.
- Minecart builder / repairer: assembles carts from actual timber and forged components.
- Haulers, carpenters, blacksmiths and builders: supply and maintain every stage of the operation.

Equipment, workshops and recipes

A prospecting toolkit should be forge-produced, drawing on existing crafting rules and physical inventories. New equipment may include picks, hammers, sample bags, sieves, assay instruments, timber supports, shovels, wheelbarrows, minecarts, axles, wheels, rail sections, track fasteners, lamps and repair parts. These are candidate recipes, not final approved quantities or metallurgy.

For each item, Claude must define realistic inputs, manufacturing workstation, labor, wear, repair, storage, transport and consumers. Do not allow one new tool to bypass existing carpentry, forging or hauling systems.

6. Stone, spoil and dirt are real resources

After surface ore is exhausted, the gray host rock is still useful stone. Quarrying must integrate with the existing construction economy, including roads, foundations, walls and masonry where supported. Excavated dirt and topsoil are separately tracked materials rather than deleted terrain.

Candidate uses for earth include grading building plots, road embankments, backfilling, defensive earthworks and landscaping. Agricultural use must distinguish suitable topsoil from poor subsoil. Storage and disposal have labor and space costs. All uses require recipes and world-system integration before implementation.

7. Mine geometry, caves and presentation

Open pits should visibly deepen through excavation, with terraced walls and a navigable spiral or switchback ramp around the perimeter. Underground workings should be enterable, not merely decorative. In the isometric presentation, overhead terrain becomes transparent, hidden or cut away when needed to keep player and NPCs visible.

Before promising terrain excavation, inspect whether the existing map, collision and navigation systems support layered elevations, altered ground, interior spaces and persistence. Hidden animal breeding refuges do not by themselves establish enterable cave support. Assess pathfinding, rendering, camera, lighting, stability, save migration and eventual multiplayer replication.

8. Autonomous settlement development

NPC settlements should evaluate resource scarcity, local geology, workforce, tools, available funds, trade opportunities and construction demand. A village with sufficient capability may train a prospector, commission equipment, investigate clues, open a quarry or expand a mine without a player issuing orders. A failed prospecting venture costs time and resources; a successful one can create new jobs and attract families, merchants and craftspeople.

Settlement advancement must emerge from demonstrated capability and sustained institutions. Track mining technology/capability, industrial workforce, infrastructure and productive capacity independently of population and political status. Skills may be retained through apprenticeships, records and living practitioners; loss of all expertise can constrain future development.

9. Economy, scarcity and balancing safeguards

- Ore bodies and stone layers remain finite. No automatic replenishment of depleted veins.
- Prospecting does not guarantee success; expansion needs time, labor, materials and investment.

- Caravans and external trade remain legitimate alternative supply paths but should not hide broken local production.
- Mine development should increase demand for timber, forge products, food, transport and construction.
- Check that extraction, waste and resource transportation conserve inventory and money.
- Track effects on housing, tools, employment, wages/prices, population, wildlife habitat and the environment.
- A settlement may fail or decline; the broader world should have credible recovery paths.

10. Release and implementation boundaries

Reserve this as a dedicated industry-development expansion, tentatively Build 17 or 18. The exact number is not locked. Do not derail or silently incorporate the full expansion into the Build 15/16 equilibrium work. Capture requirements now, conduct architecture review, and plan staged development on the normal remote GitHub staging workflow. Keep current authorized release/test gates intact.

Suggested implementation phases within the dedicated build

1. A. Audit terrain, mining, inventory, recipes, AI, pathfinding, saves and map generation; establish architecture and measurable acceptance tests.
2. B. Preserve exhausted ore outcrops as quarryable stone; add quarry chain and stone logistics.
3. C. Add natural prospecting clues, specialist profession, forge-produced toolkit, samples and assay workflow.
4. D. Add new finite deposit discovery and settlement capability levels with autonomous NPC adoption.
5. E. Add open-pit excavation, spoil handling, ramps, hauling and geological layers where architecture supports them.
6. F. Add underground/cave mining, cutaway presentation, minecarts, track construction and full industrial dependencies.
7. G. Run 100-/365-day multi-seed simulations and gameplay tests, validate saves, economic conservation and no resource spawning.

11. Acceptance criteria

- A depleted colored ore deposit remains mineable for ordinary stone without regenerating ore.
- At least one prospecting clue leads to a search area, not a guaranteed exact ore location; failures are possible.
- Prospectors and assayers use actual produced equipment and consume actual time and resources.

- The settlement can advance from surface extraction to prospecting through real prerequisites and NPC agency.
- Open-pit excavation, if implemented, removes and accounts for dirt/stone and exposes finite layers.
- Underground workings, if implemented, are enterable, readable in isometric view and navigable by NPCs.
- Minecarts, track, timber supports and tools connect to existing workshops and transport rather than appearing for free.
- Simulations demonstrate that mining employment and related industries can adapt after initial ore depletion without artificial replenishment.
- Every approved requirement is entered in requirements.json / the development ledger and cross-referenced to the Interlocking Systems Manifest.

12. Open technical decisions, not blockers for design registration

- Can the current terrain support true destructible depth, or is a staged representation required first?
- How are subsurface deposits generated, bounded, revealed and persisted across saves?
- How do settlement-level skills accumulate, spread, decay and interact with town/settlement classification?
- What exact crafting recipes, tool durability, mine hazards, construction costs and labor rates are balanced?
- What is the distinction between ordinary quarry stone, usable building blocks, spoil and topsoil?
- Which parts fit into the first dedicated expansion build versus follow-on milestones?

Directive to Claude: preserve this as the owner-approved industry direction; label the explicit candidate mechanics and level 5 as proposals pending validation. Produce an architecture assessment and implementation plan before invasive terrain changes. Continue the current Build 15/16 simulation work independently.